

Roll No.:

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B. Tech- CSE

Semester: VI

Name of the Course: Image Processing and Computer Vision

Course Code: TCS 691

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1		(20Marks)																									
(a)	What are the steps in image processing applications? Define the significance of sampling and quantization? What is the result of poor quantization effect?	CO2																									
(b)	Define Histogram equalization? Equalize the given histogram: -- <table><tr><td>Grey Level</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>No. of Pixels</td><td>790</td><td>1023</td><td>850</td><td>656</td><td>329</td><td>245</td><td>122</td><td>81</td></tr></table>	Grey Level	0	1	2	3	4	5	6	7	No. of Pixels	790	1023	850	656	329	245	122	81	CO1							
Grey Level	0	1	2	3	4	5	6	7																			
No. of Pixels	790	1023	850	656	329	245	122	81																			
(c)	Differentiate between image enhancement and image restoration.	CO1																									
Q2																											
(a)	A source emits letters from alphabet $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5)$ with probabilities $P(\alpha_1) = 0.2$, $P(\alpha_2) = 0.4$, $P(\alpha_3) = 0.2$, $P(\alpha_4) = 0.1$ and $P(\alpha_5) = 0.1$. Find a Huffman code for this source? Find the average length of this code and its redundancy?	CO3																									
(b)	What are the multiple ways of finding dilation and erosion for binary images? Consider the following image A and structuring element B. <table><tr><td>7</td><td>8</td><td>2</td><td>4</td></tr><tr><td>6</td><td>4</td><td>3</td><td>3</td></tr><tr><td>7</td><td>3</td><td>6</td><td>6</td></tr><tr><td>4</td><td>4</td><td>2</td><td>3</td></tr></table> A <table><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>2</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table> B	7	8	2	4	6	4	3	3	7	3	6	6	4	4	2	3	1	1	1	1	2	1	1	1	1	CO5
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1	2	1																									
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(c)	Discuss various spatial domain image enhancement techniques in detail?	CO2																									

Q3																																																																		
(a)	Draw and explain the architecture of Convolutional Neural Network? Explain the concept of convolutional operation in CNN?	CO2																																																																
(b)	Explain the stages of edge detection algorithm? How are they present in edge operators? Consider the following image: - $A =$<table><tr><td>1</td><td>2</td><td>5</td></tr><tr><td>5</td><td>5</td><td>5</td></tr><tr><td>5</td><td>3</td><td>2</td></tr></table> Show the output of any edge detection algorithm?	1	2	5	5	5	5	5	3	2	CO6																																																							
1	2	5																																																																
5	5	5																																																																
5	3	2																																																																
(c)	Differentiate between lossless and lossy compression algorithms.	CO3																																																																
Q4																																																																		
(a)	Explain the core concepts and architecture of the You Only Look Once (YOLO) algorithm for object detection.	CO4																																																																
(b)	Elaborate the concept of Otsu thresholding. Consider the histogram of a 3-bit image as shown below: <table><tr><td>Gray Level</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>No. of Pixels</td><td>3</td><td>2</td><td>1</td><td>3</td><td>2</td><td>3</td><td>2</td><td>1</td></tr></table> Find out the optimal thresholding value using Otsu.	Gray Level	0	1	2	3	4	5	6	7	No. of Pixels	3	2	1	3	2	3	2	1	CO6																																														
Gray Level	0	1	2	3	4	5	6	7																																																										
No. of Pixels	3	2	1	3	2	3	2	1																																																										
(c)	Discuss K- means clustering with suitable example. Discuss advantages and disadvantages of K-means clustering.	CO5																																																																
Q5																																																																		
(a)	Draw and explain the architecture of Generative Adversarial Network. How you can calculate the cost function for Generative Adversarial Network. Also, explain the drawbacks of Generative Adversarial Network	CO2																																																																
(b)	Apply low pass median filtering process on following image segment: <table><tr><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td></tr><tr><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td></tr><tr><td>10</td><td>250</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td></tr><tr><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td></tr><tr><td>50</td><td>50</td><td>50</td><td>50</td><td>50</td><td>50</td><td>50</td><td>50</td></tr><tr><td>50</td><td>50</td><td>80</td><td>50</td><td>50</td><td>50</td><td>50</td><td>50</td></tr><tr><td>50</td><td>50</td><td>50</td><td>50</td><td>50</td><td>20</td><td>50</td><td>50</td></tr><tr><td>50</td><td>50</td><td>50</td><td>50</td><td>50</td><td>50</td><td>50</td><td>50</td></tr></table>	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	250	10	10	10	10	10	10	10	10	10	10	10	10	10	10	50	50	50	50	50	50	50	50	50	50	80	50	50	50	50	50	50	50	50	50	50	20	50	50	50	50	50	50	50	50	50	50	CO5
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(c)	Discuss various applications of image processing and computer vision in detail.	CO2																																																																